CLOSED IMMERSIONS DESCEND TO A FINITE LEVEL IN THE DESCENT SITUATION

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Editorial note. No omissions in the counted span: the reduction is worked; the algebra core is a precise citation of a proved lemma.

Allowed standard prerequisites (editorial).

- Colimit surjective. Algebra, Lemma algebra-lemma-colimit-surjective
- Descend affine fp. Limits, Lemma lemma-descend-affine-finite-presentation (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SE: sources/stacks-chow/tags.txt, source lines 2335–2335. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZN: sources/stacks-chow/tags.txt, source lines 2600–2600. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07RH: sources/stacks-chow/tags.txt, source lines 9867–9867. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If*

(1) *f is a closed immersion, and*
(2) *f_0 is locally of finite type,*

then there exists an $i \geq 0$ such that f_i is a closed immersion.

Proof. A closed immersion is affine, see Morphisms, Lemma 01SE. Hence by Lemma 01ZN above after increasing 0 we may assume that f_0 is affine. By writing Y_0 as a finite union of affines we reduce to proving the result when X_0 and Y_0 are affine and map into a common affine $W \subset S_0$. The corresponding algebra statement is a consequence of Algebra, Lemma 07RH. □

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